

■ A.3 Mathematical Operations and Objects

■ A.3.1 Vector Analysis

Vector analysis is used in many areas of applied mathematics. This section gives part of a simple package for doing vector analysis in three dimensions, with some specific coordinate systems.

```

Coordinates[Cartesian] ^= {x, y, z}
ScaleFactors[Cartesian] ^= {1, 1, 1}

Coordinates[Cylindrical] ^= {r, phi, z}
ScaleFactors[Cylindrical] ^= {1, r, 1}

Coordinates[Spherical] ^= {r, theta, phi}
ScaleFactors[Spherical] ^= {1, r, r Sin[theta]}

VolumeElement[system_:$CoordinateSystem] :=
  Block[{i, s=ScaleFactors[system], c=Coordinates[system]},
    Product[s[[i]] Dt[c[[i]]], {i, 3}]
  ]

Grad[f_, system_:$CoordinateSystem] :=
  Block[{i, s=ScaleFactors[system], c=Coordinates[system]},
    Table[ D[ f, c[[i]] ] / s[[i]], {i, 3}]
  ]

Div[f_?VectorQ, system_:$CoordinateSystem] :=
  Block[{i, s=ScaleFactors[system], c=Coordinates[system], v},
    v = Product[s[[i]], {i, 3}] ;
    Sum[ D[f[[i]] v/s[[i]], c[[i]]], {i, 3} ] / v
  ]

```

Some operations for vector analysis.

Read in the vector analysis package.

```
In[1]:= <<VectorAnalysis.m
```

Here is the volume element in spherical coordinates.

```
In[2]:= VolumeElement[Spherical]
Out[2]= r2 Dt[phi] Dt[r] Dt[theta] Sin[theta]
```

Here is the gradient of the function $r^2 \cos(\theta) \sin(\phi)$, in spherical coordinates.

```
In[3]:= Grad[ r^2 Cos[theta], Spherical ]
Out[3]= {2 r Cos[theta], -(r Sin[theta]), 0}
```

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